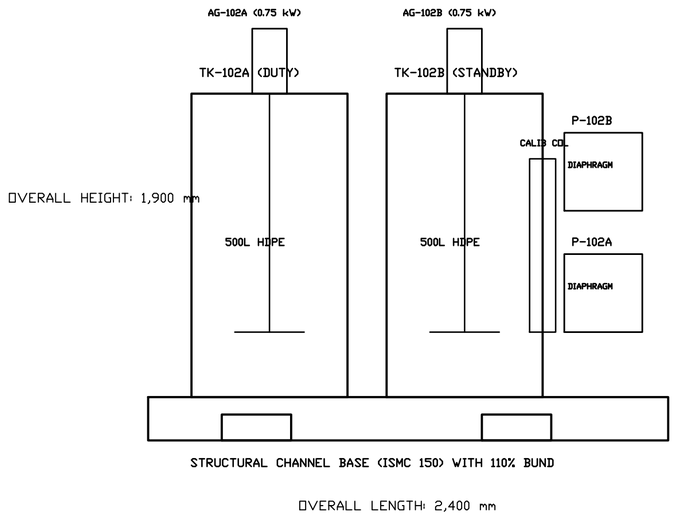



## CHEMICAL REAGENT DOSING SKID SPECIFICATION & BILL OF MATERIALS (SK-100)

| TAG | EQUIPMENT / COMPONENT | CAPACITY / RATING | METALLURGY | PROCESS SERVICE & SPECIFICATION |
|---|---|---|---|---|
| TK-102A/B | Dosing Batch Tanks (2x) | 500 L (800 Dia x 1100) | Food-Grade HDPE | Dual batch prep (duty/standby), ultrasonic LT-102 |
| AG-102A/B | High-Shear Agitators (2x) | 0.75 kW @ 1440 RPM | AISI SS316L | Rapid salt dissolution (bicarbonate/citrate salts) |
| P-102A/B | Metering Pumps (2x) | 14.06 L/h (5–50 L/h) | PVDF / PTFE Diaph | Positive displacement diaphragm, 10:1 turndown VFD |
| PSV-102A/B | Safety Relief Valves | Set @ 4.0 bar g | PVDF Liquid End | Full bypass recirculation back to preparation tank |
| PD-102 | Pulsation Dampener | 0.5 L Bladder Type | PVDF / Viton | Smooths discharge pulsation to < +/-2.5% variation |
| CC-102 | Calibration Column | 1,000 mL Cylinder | Borosilicate Glass | 3-Way valve draw-down flow verification cylinder |
| PIP-102 | Skid Piping Network | 1" NB Schedule 10S | AISI SS316L | Sanitary tri-clamp spools, zero dead-leg geometry |
| BD-102 | Spill Containment Bund | 600 L (110% Capacity) | FRP Lined CS | Containment tray with optical liquid leak sensor LS-102 |
| V-102A–H | Sanitary Ball Valves | 1" NB 150# Tri-Clamp | AISI SS316L / PTFE | Cavity filled, 3-piece full bore isolation valves |
| CP-102 | Local Control Panel | IP65 Wall Mount | AISI SS304 Casing | VFD drives, manual stroke control, E-Stop push button |
| SKID FRAME | Structural Base Skid | 2,400 x 1,400 x 1,900 | AISI SS304 Box Sec | Epoxy-grouted leveling feet, seismic anchor lugs |

DESIGN COMPLIANCE: FDA 21 CFR 177.1520 food-contact certified; CGMP sanitary construction.

AUTOMATION INTEGRATION: Modbus TCP communication to central SCADA; 4–20mA pacing proportional to dal flow.

DOSING ACCURACY: Micro-meter stroke adjustment delivering +/- 0.5% volumetric repeatability across range.

ENVIRONMENTAL SAFETY: 110% bund capacity prevents chemical runoff; interlocked with plant emergency shutdown.

## SKID NOZZLE SCHEDULE & UTILITY INTERFACE SCHEDULE

| NOZZLE | SERVICE DESCRIPTION | SIZE | RATING | FACING | METALLURGY | CONNECTED TERMINATION |
|---|---|---|---|---|---|---|
| N1 | Soft Water Supply Infeed | 1.5" NB | ANSI 150# | Flat Face | AISI SS316L | WTP-01 Header @ 2.5 bar g |
| N2 | Chemical Dosing Discharge | 1.0" NB | ANSI 150# | Raised Face | AISI SS316L | Soak Battery TK-101 Feed Spool |
| N3 | Containment Bund Drain | 2.0" NB | ANSI 150# | Flat Face | AISI SS304 | ETP Equalization Pit TK-104 |
| N4 | Tank Overflow Vent | 1.5" NB | ANSI 150# | Atmospheric | Food-Grade HDPE | Gravity drain to internal skid bund |
| E1 | Main Electrical Power Feed | 4-Core 4 mm2 | 415V / 50Hz | Glanded | Armoured Cu | MCC-01 Feeder Feat-04 (3.5 kW connected) |
| I1 | Automation Signal Trunk | 12-Pair Shielded | 4–20mA / DI | Glanded | Tinned Cu | Remote I/O Panel RIO-02 (Modbus TCP) |

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GA – CHEMICAL DOSING SKID SK-100
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

| DWG NO: DWG-FIN-15T-009 | REV: V8.0 | SHEET: 9 OF 13 |
|---|---|---|
| SCALE: AS SHOWN DATE: 2026-10-17 | STATUS: PRELIMINARY | GATE: G-A HOLD |

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